VIDYAVARDHINI'S COLLEGE OF SCIENCE (VARTAK COLLEGE), VASAI ROAD (W)
Department of Computer Science • Affiliated with University of Mumbai • Academic Year 2026–2027

COMMUNITY ENGAGEMENT PROJECT (NEP 2020) PROJECT OVERVIEW & EVALUATION HANDBOOK

PROJECT TITLE: **Local Biodiversity Mapping of Byculla Zoo & Surrounding Green Spaces**	**INTERNAL GUIDE / MENTOR:** **Prof. Shweta Iyer** **Dept. of Computer Science**
CLASS & WEIGHTAGE: T.Y. B.Sc. CS — Semester V (NEP 2020: 2 Credits)	**DEPLOYED WEB-GIS APPLICATION:** **https://biomap-byculla.netlify.app**

☐ 1. What is Our Project About? (The Core Summary)

The 30-Second Elevator Pitch

Our project is an empirical ecological audit and digital mapping platform for the free-living, non-captive biodiversity of Mumbai's 53-acre Byculla Zoo (Veermata Jijabai Bhosale Udyan / Ranibagh). Because over 78% of visitors focus exclusively on caged zoo exhibits (like penguins and tigers), the park's rich botanical heritage (280+ tree species) and wild avifauna are unmapped and overlooked.

To solve this, we collected on-site GPS coordinates, quantified a 26.3 dBA acoustic attenuation shield delivered by the canopy, surveyed 84 community members, and engineered 'BioMap' (https://biomap-byculla.netlify.app) — a zero-cost, open-source mobile Web-GIS platform providing real-time GPS tracking and bilingual (Marathi/English) ecological profiles.

□ 2. The Problem Statement & Civic Motivation

1. The 53-Acre Heritage Sanctuary: Established in 1861, Ranibagh is South Mumbai's largest contiguous botanical sanctuary, harboring century-old trees and native birdlife.

2. The 'Captive-Species Bias': Public attention is heavily skewed towards caged animals. Botanical trees and wild avifauna receive zero interpretation on physical placards.

3. Arbitrary Pre-Monsoon Tree Lopping: Municipal contractors routinely lop branches prior to the monsoon, destroying active bird nesting cavities without prior ecological audits.

4. Acoustic Pollution Vulnerability: Heavy vehicular traffic along the perimeter (Dr. Ambedkar Road) regularly exceeds 78 dBA, creating severe acoustic stress for vocalizing avifauna.

5. The Digital Literacy Void: No interactive digital map or vernacular mobile guide existed for citizens to explore native flora and fauna on their smartphones.

□ 3. Study Area & 4-Zone Ecological Stratification

We stratified the 53-acre park into 4 distinct micro-habitats to audit ecological variation and sound levels:

Zone	Habitat Character	GPS Coordinates	Dominant Species Audited
Zone 1	Botanical Grove (Heritage Arboretum)	18°58'38"N, 72°49'51"E	African Baobab, Ceylon Ebony, Coppersmith Barbet (Noise: 52.1 dBA)
Zone 2	Freshwater Wetland & Lotus Lake	18°58'42"N, 72°49'56"E	Black-crowned Night Heron, Common Moorhen, Odonata (Noise: 54.8 dBA)
Zone 3	Promenade Shrublands & Butterfly Lawns	18°58'35"N, 72°49'48"E	Common Mormon Butterfly, Bluebottle, Malabar Nut/Adulsa (Noise: 61.2 dBA)

Zone 4	Perimeter Transition Buffer Wall	18°58'30"N, 72°49'44"E	Black Kite, House Crow, Common Myna (Noise: 78.4 dBA — High Stress)

Instrumentation Stack: Garmin eTrex GPS & Mobile GNSS (±3m accuracy), Digital Sound Level Meter (Type 2, A-weighting, 15-min intervals), Nikon 70-300mm telephoto optics, and fixed 100m transect census walks (06:30–09:30 AM).

☐ 4. Key Scientific & Quantitative Discoveries

⍰ Headline Finding 1: The '26.3 dBA Acoustic Shield' Effect

- Ambient traffic noise at the outer boundary (Zone 4) reaches 78.4 dBA.
- Deep inside the Botanical Grove (Zone 1), sound levels plummet to 52.1 dBA.
- The multi-tier arboreal tree canopy acts as a massive acoustic sponge, absorbing 26.3 dBA of noise. This creates a critical sonic refuge allowing birds to vocalize, mate, and nest without acoustic interference.

⍰ Headline Finding 2: 81% Higher Shannon-Wiener Diversity in Core Grove

- We calculated the Shannon-Wiener Diversity Index: H' = - Σ (pi × ln pi) and Pielou's Evenness J' = H' / ln(S).
- Zone 1 achieved H' = 2.68 (24 taxa, J' = 0.84), compared to Zone 4 which had H' = 1.48 (only 6 edge-tolerant taxa, J' = 0.65).
- Validates that canopy continuity and aquatic features are indispensable for urban ecological resilience.

☐ 5. Community Survey & Perception Analysis (N = 84 Citizens)

In strict alignment with NEP 2020 social engagement criteria, we surveyed 84 diverse stakeholders (morning walkers, school teachers, students, tourists, and BMC caretakers):

- **Habitat Clustering (Q3):** 56.5% of free-living wildlife encounters occur strictly in the inner Botanical Grove and Lake.
- **Greatest Perceived Threat (Q4):** 36.5% cite pre-monsoon tree trimming, followed by 27.0% citing perimeter traffic noise.
- **Biodiversity Literacy Deficit (Q5):** 72.0% cite the absence of digital maps and signage restricted to cages as their main barrier.
- **Validation of Digital Solution (Q6 & Q7):** 86.7% confirmed an interactive mobile web app would enhance their visit, and 83.3% stated they would actively use it.
- **Demand for Tree QR Codes (Q10):** 87.0% consider QR-code placards on heritage trees linking to species profiles highly beneficial.

☐ 6. Technological Solution: BioMap Web-GIS Platform

To translate our research into a permanent digital public good, we built and deployed BioMap:

- **Live URL & Architecture:** Fully deployed and accessible on mobile browsers at https://biomap-byculla.netlify.app with zero download friction.
- **Real-Time Mobile GPS Tracking:** HTML5 Geolocation API displays the visitor's real-time position on 53 acres of garden trails.
- **Interactive 4-Zone Overlays:** Semi-transparent GeoJSON polygon boundaries communicating microclimate, sound levels, and species counts.
- **Bilingual Ecological Cards:** Profiles include Linnaean hierarchy, IUCN status, flowering seasons, and Marathi vernacular descriptions (□ □ □ □ □ □ □ □□□□ □ □).
- **Zero-Cost Open-Source Stack:** Leaflet.js + OpenStreetMap + Vanilla CSS3 — zero ongoing database or municipal server costs, making it easily replicable for other municipal parks.

□ 7. Actionable Recommendations (Three-Tier Blueprint)

- **Tier 1: Park Level (Immediate / 1–3 Months):** Laser-etched QR code placards on 42 audited heritage trees; mandatory pre-pruning bird-nest inspections between February and June; floating reed resting islands in Lotus Lake.
- **Tier 2: BMC Municipal Level (Policy / 6–12 Months):** Integrate BioMap's GIS dataset into BMC's statutory People's Biodiversity Register (PBR); plant a 50m dense acoustic green buffer along Dr. Ambedkar Road.

• **Tier 3: Citizen Science & Academia (Ongoing):** Launch university Bio-Blitz weekends for CS/IT and science undergraduates; enable citizen sighting submissions with botanist verification.

☐ 8. What Evaluators Focus On & Viva Defense Cheat Sheet

5 Key Criteria Evaluators Grade in Your Report PDF

1. Academic Structure: Title page, certificate signed by Prof. Shweta Iyer, declaration, 5 NEP chapters, references, diary.
2. Interdisciplinary Rigor: Bridging Computer Science (Web-GIS) with municipal urban ecology (not just basic coding).
3. Field Authenticity: On-site GPS coordinates, geotagged species photos, and 42+ taxa census.
4. Community Evidence: N = 84 citizen surveys with quantitative charts and actionable findings.
5. Measurable Findings: 26.3 dBA acoustic damping, Shannon Index (H' = 2.68), and a deployed live website.

Viva Question	Winning 1-Minute Model Answer
Why did you choose Byculla Zoo for a CS project?	Byculla Zoo is South Mumbai's largest botanical sanctuary (53 acres, 280+ tree species), but public awareness is trapped in a 'captive-species bias' focusing only on caged animals. We applied Computer Science & GIS to map the neglected free-living wildlife and provide an accessible digital public tool.
What is your project's biggest scientific finding?	The multi-tier botanical canopy acts as an acoustic shield, dampening perimeter city traffic noise from 78.4 dBA down to 52.1 dBA (a 26.3 dBA reduction), preserving an 81% higher Shannon biodiversity index (H' = 2.68 vs. 1.48) in the park center.
What technology did you use for BioMap?	We used Leaflet.js, OpenStreetMap, HTML5 Geolocation, and GeoJSON polygon layers deployed on Netlify. It requires zero server licensing costs and runs instantly on any smartphone browser without app installation.
How did you incorporate community engagement?	We surveyed 84 diverse citizens (morning walkers, students, caretakers). 86.7% confirmed the need for a digital guide, and their feedback shaped our bilingual (Marathi/English) interface and QR-code recommendation.
Who guided your research?	The project was mentored and supervised by Prof. Shweta Iyer, Department of Computer Science, Vartak College, through 10 formal supervisory meetings documented in our Guide Interaction Diary.